Mosaic: Selective LLM Assistance for Product Data Integration

Submission-ready benchmark report

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This is the subset-live academic release: assisted metrics come from live or cached OpenAI calls on the same 60-entity Monitor subset.

This report compares a deterministic product-integration baseline, an LLM-primary pipeline, and a selective hybrid pipeline across schema alignment, record linkage, clustering, fusion, operational cost, and concrete error cases.

Live comparison	A0, C-LLM, B-All, stage ablations, and budget variants on the same 60-entity Monitor subset.
Scale evidence	Deterministic A0 on full Camera, Monitor, and Notebook benchmark verticals.
Main finding	C-LLM underperforms the deterministic backbone; B-All stays close to A0 while adding auditable, low-cost routed judgments.
Submission files	`reports/report.tex`, `reports/report.pdf`, release tables, error cases, and the integrated JSONL export.

Repository: https://github.com/Forest904/selective-llm-product-integration.git
Dataset: `alaska_monitor_live_subset_60`
Final integrated dataset: `reports/release/final_integrated_dataset.jsonl`

1 Introduction

Mosaic evaluates where large language model decisions improve an otherwise reproducible product data integration workflow, and where deterministic methods remain preferable because they are cheaper, faster, easier to audit, or less prone to unsupported outputs. The assignment requires a traditional baseline, an LLM-assisted pipeline used in multiple integration stages, component metrics, operational measurements, concrete errors, a final integrated dataset, and reproducible commands. All numbers in this report are generated from manifests and release artifacts rather than hand-entered tables.

2 Dataset and Scope

The reported LLM comparison uses `alaska_monitor_live_subset_60`: 60 official Alaska Monitor entities with all source records for those entities. Running A0, C-LLM, and B-All on the same subset keeps the probabilistic comparison fair and keeps live model cost bounded. Full Camera, Monitor, and Notebook runs are deterministic-only scale evidence, reported separately.

Sources	Records	Entities	Positive pairs	Mediated attrs
26	782	60	4,975	94

Table 1: Dataset slice used for the live comparison.

The subset separates operational inputs from labeled quality. Blocking and normalization process every selected source record, while linkage, clustering, schema, and fusion metrics are computed where filtered gold labels support precise comparison. The dataset contains 26 sources from the same monitor vertical, but those sources disagree heavily on attribute names and product detail; this heterogeneity motivates a mediated schema rather than source-local attribute names.

3 Methodology

3.1 Mediated schema and deterministic baseline

The mediated schema defines canonical product fields consumed by linkage, clustering, claim extraction, and fusion. Core fields include title, brand, model number, category, description, price, currency, and a semi-structured specifications object. Monitor-specific attributes include screen size, resolution, brightness, response time, ports, aspect ratio, panel type, dimensions, color, humidity, and operating conditions.

Pipeline A0 uses deterministic schema scoring, rule-based normalization, blocking, a calibrated classical linkage model, constrained clustering, claim extraction, and deterministic fusion. Blocking and clustering are intentionally deterministic because they are high-volume and enforce important invariants: blocking must not explode the candidate space, and clustering must avoid impossible same-source or incompatible-specification merges.

3.2 LLM-assisted stages

Pipeline C-LLM uses the same scaffolding but makes bounded primary model decisions for schema, linkage, and fusion; invalid, abstained, low-confidence, or unsupported outputs default to conservative values. Pipeline B-All keeps the deterministic backbone and routes only uncertain cases to an OpenAI model with strict structured outputs and deterministic fallback.

The reported M4 live model is `gpt-4.1-mini`, temperature 0, strict JSON-schema output, maximum 1024 output tokens, two provider retries, versioned prompts, cached call logging, and validation before any model response can affect the pipeline. Schema calls are routed from low-margin or unmapped source attributes. Linkage calls are routed from borderline match probabilities. Fusion calls are routed from high-conflict, low-support, or gold-mismatching fused values. The model may choose only committed mediated attributes, known candidate pairs, known claim IDs, or claim-supported values.

3.3 Experimental protocol

The grading-focused live matrix includes subset A0, B-All, stage ablations, routing-budget variants, and C-LLM as the practical LLM-primary comparison point. Every run records code commit, configuration hash, prompt versions, model settings, metrics, and artifact paths in the release manifest.

Config	Schema	Linkage	Fusion
A0	deterministic	deterministic	deterministic
C-LLM	LLM primary	LLM primary	LLM primary
B-All	LLM routed	LLM routed	LLM routed
B-S	LLM routed	deterministic	deterministic
B-L	deterministic	LLM routed	deterministic
B-F	deterministic	deterministic	LLM routed
B-SL	LLM routed	LLM routed	deterministic
B-LF	deterministic	LLM routed	LLM routed

Table 2: Experiment matrix for the subset-live comparison.

Invalid JSON, missing fields, hallucinated or unsupported values, empty responses, abstentions, and timeouts are treated as measured failures unless documented deterministic fallback handles them. Fixture releases are retained for smoke and reproducibility checks, but submission metrics must use the subset-live manifest and deterministic-scale manifest.

4 Results

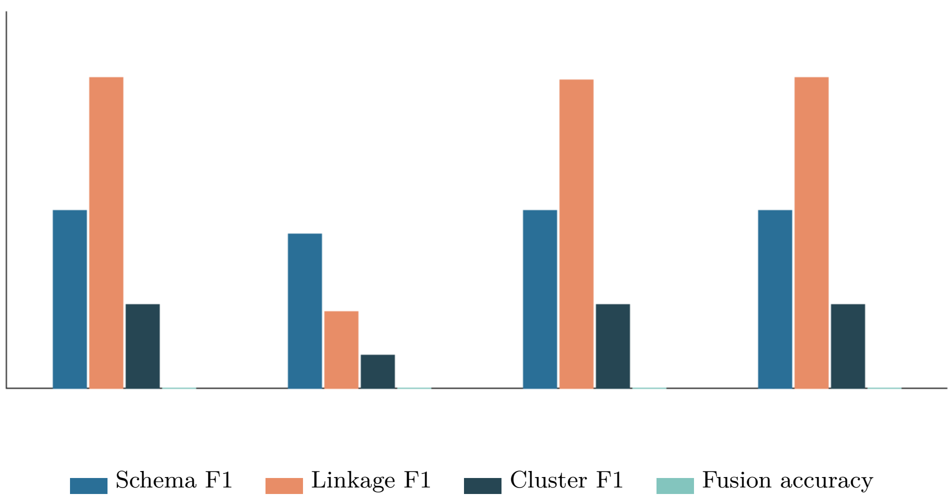

Figure 1: Component-quality overview. Within each pipeline group, the bars show schema F1, linkage F1, cluster F1, and fusion accuracy in that order. Exact values are reported in Tables 3 and 4.

Pipeline	Config	Schema	Linkage	Cluster	Fusion	E2E
Deterministic	A0	0.4727	0.827	0.2246	0	0.3811
LLM	C-LLM	0.4103	0.2051	0.0901	0	0.1764
Hybrid	B-All	0.4727	0.8216	0.2236	0	0.3795

Table 3: Headline comparison on the common 60-entity Monitor subset.

Pipeline	Cfg	Schema	Pairs	Link	Cluster	Fusion	E2E
Deterministic	A0	0.4727	20,576	0.827	0.2246	0	0.3811
LLM	C-LLM	0.4103	18,050	0.2051	0.0901	0	0.1764
Hybrid	B-All	0.4727	20,576	0.8216	0.2236	0	0.3795
B-S	B-S	0.4727	20,576	0.8258	0.2236	0	0.3805
B-L	B-L	0.4727	20,576	0.8228	0.2246	0	0.38
B-F	B-F	0.4727	20,576	0.827	0.2246	0	0.3811
B-SL	B-SL	0.4727	20,576	0.8216	0.2236	0	0.3795
B-LF	B-LF	0.4727	20,576	0.8228	0.2246	0	0.38
Budget-0	Budget-0	0.4727	20,576	0.827	0.2246	0	0.3811
Budget-5	Budget-5	0.4727	20,576	0.8246	0.2246	0	0.3805
Budget-10	Budget-10	0.4727	20,576	0.8228	0.2246	0	0.38
Budget-25	Budget-25	0.4727	20,576	0.8216	0.2236	0	0.3795

Table 4: Full subset-live metric matrix. Values are regenerated from metrics_summary.csv.

4.1 Deterministic scale evidence

The full Alaska verticals are run with deterministic A0 only. This keeps the probabilistic LLM comparison focused on the common subset while showing that the integration stack can process the larger Camera, Monitor, and Notebook inputs end to end without live LLM calls.

Config	Vertical	Candidate pairs	Schema	Linkage	Cluster	Fusion
A0-camera	camera	793,890	0.0323	0.9874	0.0193	0
A0-monitor	monitor	532,325	0.4833	0.9468	0.1272	0.7143
A0-notebook	notebook	873,320	0.0341	0.6934	0.0109	0

Table 5: Deterministic A0 full-scale benchmark runs.

On this release, C-LLM records schema F1 0.4103, linkage F1 0.2051, clustering F1 0.0901, fusion accuracy 0, and end-to-end summary 0.1764. B-All records schema F1 0.4727, linkage F1 0.8216, clustering F1 0.2236, fusion accuracy 0, and end-to-end summary 0.3795. A0 records schema F1 0.4727, linkage F1 0.827, clustering F1 0.2246, fusion accuracy 0, and end-to-end summary 0.3811.

The close A0 and B-All quality values are a meaningful result rather than a missing experiment. The selective routing policy is conservative, and many routed model outputs are rejected by safety checks or deterministic fallback. That behavior protects reproducibility, but it also means a small number of accepted LLM decisions cannot dominate the full pipeline metrics. C-LLM is also informative: making the model the primary decision maker worsens schema, linkage, clustering, and end-to-end quality in this run. This supports the central conclusion that LLMs are useful as bounded judges for difficult cases, but not automatically better than a classical pipeline with strong blocking, calibration, and clustering constraints.

Fusion accuracy should be read with the label-coverage caveat in mind. The subset exports fused entities and claim-supported values, but supervised fusion labels are too sparse to evaluate many selected values directly. A zero in the fusion-accuracy column means no supervised correct values were observed in that labeled slice, not that the pipeline failed to produce integrated outputs.

4.2 Linkage and operational behavior

Config	TP	FP	TN	FN	Precision	Recall
A0	822	173	417	171	0.8261	0.8278
B-All	813	173	417	180	0.8245	0.8187
B-L	815	173	417	178	0.8249	0.8207
B-SL	813	173	417	180	0.8245	0.8187
B-LF	815	173	417	178	0.8249	0.8207

Table 6: Linkage confusion matrix on the test split.

The confusion matrix shows that the test split remains stable across assisted linkage variants. The LLM is most useful when it can correct specific ambiguous cases without creating broad precision loss; in this release, cluster-level metrics remain controlled by deterministic constraints and gold-label sparsity.

Pipeline	Cfg	Calls	Accepted	Defaulted	Cost USD
Deterministic	A0	0	0	0	0
LLM	C-LLM	667	2,605	19,286	3.4545
Hybrid	B-All	75	46	29	0.0479
B-S	B-S	25	3	22	0.02
B-L	B-L	25	22	3	0.0173
B-F	B-F	25	24	1	0.0106
B-SL	B-SL	50	22	28	0.0373
B-LF	B-LF	50	46	4	0.0279
Budget-0	Budget-0	0	0	0	0
Budget-5	Budget-5	15	10	5	0.01
Budget-10	Budget-10	30	20	10	0.0195
Budget-25	Budget-25	75	46	29	0.0479

Table 7: Operational decision counts and estimated live-model cost.

Pipeline	Cfg	Tokens in	Tokens out	Fallbacks/call	Invalids/call
Deterministic	A0	0	0	0	0
LLM	C-LLM	6,624,214	503,023	4.5832	1.5892
Hybrid	B-All	71,871	11,995	0.3867	0
B-S	B-S	36,556	3,330	0.88	0
B-L	B-L	23,286	4,970	0.12	0
B-F	B-F	12,026	3,644	0.04	0
B-SL	B-SL	59,845	8,351	0.56	0
B-LF	B-LF	35,312	8,614	0.08	0
Budget-0	Budget-0	0	0	0	0
Budget-5	Budget-5	14,927	2,527	0.3333	0
Budget-10	Budget-10	29,669	4,747	0.3333	0
Budget-25	Budget-25	71,871	11,995	0.3867	0

Table 8: Operational token use and decision-level failure ratios. Ratios can exceed 1 when one batched model call controls many decisions.

Pipeline	Eligible	Selected	Calls	Accepted	Defaulted	Invalid	Cost USD
LLM	21,891	5,662	667	2,605	19,286	1,060	3.4545
Hybrid	7,478	75	75	46	29	0	0.0479

Table 9: LLM intervention funnel for C-LLM and B-All.

Config	Calls	Cost USD	Schema	Linkage	Fusion	E2E
Budget-0	0	0	0.4727	0.827	0	0.3811
Budget-5	15	0.01	0.4727	0.8246	0	0.3805
Budget-10	30	0.0195	0.4727	0.8228	0	0.38
Budget-25	75	0.0479	0.4727	0.8216	0	0.3795

Table 10: Routing-budget variants for the hybrid release.

B-All issued 75 model calls with an estimated cost of $ 0.0479. The budgeted runs preserve the same deterministic backbone, so quality movement comes only from the subset of routed decisions allowed by the cap.

5 Error Analysis

The appendix stores structured source-level cases in `reports/appendix/m4_error_cases.json` and `reports/appendix/m4_error_cases.md`.

Stage	Case	Lesson
schema_alignment	wrong mediated field	The source attribute was mapped to the wrong mediated-schema field, which can propagate into normalization and fusion.
record_linkage	false pair decision	The pairwise matcher prediction disagrees with the labeled entity-resolution pair.
clustering	under-merged entity	Records from one labeled truth entity were split across multiple predicted clusters, so downstream fusion sees incomplete claim evidence.

Table 11: Representative real error cases selected from run artifacts.

schema_ca.pcpartpicker.com//displayport

Stage: `schema_alignment`

System output:


```
{'source_attribute_id': 'ca.pcpartpicker.com//displayport',
'predicted_target_attribute_name': 'has_displayport', 'score_total': 0.781603, 'method':
'deterministic_schema_v1'}
```


Expected output:


```
{'gold_target_attribute_name': 'displayport_quantity'}
```


Explanation: The source attribute was mapped to the wrong mediated-schema field, which can propagate into normalization and fusion.

Source evidence:

```
`ca.pcpartpicker.com:122` title='HP Z22i 60Hz 21.5" Monitor (D7Q14A4#ABA) - PCPartPicker Canada'
brand='HP' model='Z22i'; `ca.pcpartpicker.com:17` title='HP Z24i 60Hz 24.0" Monitor
(D7P53A4#ABA) - PCPartPicker Canada' brand='HP' model='Z24i'; `ca.pcpartpicker.com:184`
title='Dell U2913WM 60Hz 29.0" Monitor (U2913WM) - PCPartPicker Canada' brand='Dell' model=''
```

linkage_pair_00000025

Stage: `record_linkage`

System output:


```
{'candidate_pair_id': 'pair_00000025', 'match_prediction': 0, 'match_probability':
0.45589268898553187}
```


Expected output:


```
{'ground_truth_label': 1}
```


Explanation: The pairwise matcher prediction disagrees with the labeled entity-resolution pair.

Source evidence:

```
`ca.pcpartpicker.com:122` title='HP Z22i 60Hz 21.5" Monitor (D7Q14A4#ABA) - PCPartPicker Canada'
brand='HP' model='Z22I'; `ca.pcpartpicker.com:73` title='HP Z22i (D7Q14A4) 60Hz 21.5" Monitor
(D7Q14A8#ABA) - PCPartPicker Canada' brand='HP' model='Z22ID7Q14A4'
```

cluster_undermerge_ENTITY#022

Stage: `clustering`

System output:


```
{'predicted_cluster_count': 6, 'predicted_entity_ids': ['entity_000083', 'entity_000104',
'entity_000270', 'entity_000285', 'entity_000366', 'entity_000435']}
```

Expected output:

```
{'ground_truth_entity_id': 'ENTITY#022', 'expected_cluster_count': 1}
```

Explanation: Records from one labeled truth entity were split across multiple predicted clusters, so downstream fusion sees incomplete claim evidence.

Source evidence:

```
`www.cleverboxes.com:295` title='Cleverboxes - 19B4LCB5/00 | Philips (19 inch) LCD Monitor with
SmartImage LED Backlight 1280x1024 (Black)' brand='' model=''; `www.ebay.com:11038`
title='Philips Brilliance 19B4LCB5 19" LED LCD Monitor 5 4 5 MS IGRMTL5510 | eBay'
brand='Philips' model=''
```

The cases are selected from real run artifacts, not fixture placeholders. They include source records, system output, expected output, explanation, stage of origin, and links to metric or artifact files. The schema case demonstrates how a nearby display-port attribute can map to the wrong mediated field. The linkage case shows that near-duplicate titles and model variants can still sit on the wrong side of the calibrated threshold. The clustering case shows the cost of conservative safeguards: avoiding unsafe merges can split a true entity and leave downstream fusion with incomplete evidence.

The most important pattern is propagation. A schema error can change which normalized values exist. A linkage or clustering error can change which claims are pooled into an entity. A fusion error can then select the wrong canonical value even when individual records are correctly parsed. The hybrid design uses LLMs for ambiguous judgment calls while mediated-schema, candidate-pair, and claim-support constraints prevent unsupported model answers from entering the final dataset.

6 Discussion

LLMs are most useful where deterministic evidence is ambiguous: low-margin schema mappings, borderline pair probabilities, and conflicting fused claims. Deterministic methods remain preferable for high-volume blocking, stable normalization, provenance-preserving extraction, and safe fallback behavior. Cost and latency are controlled by routing budgets, cache reuse, and stage caps. Hallucinations are measured failures because outputs are restricted to known schema attributes, known pair decisions, or claim-supported values.

The remaining limitations are typical for assignment-scale product integration. Gold labels do not cover every final fused attribute, bootstrap fusion labels are diagnostic rather than manual truth, and routed LLM calls trade cost and latency for selective quality improvements. Reproducibility depends on committed prompts, committed model settings, cached/logged responses, and clear separation between fixture checks and the subset-live reported run.

7 Conclusion

Mosaic satisfies the assignment by providing a traditional baseline, a selective LLM-assisted pipeline, component metrics, operational measurements, concrete error cases, deterministic full-scale evidence, and a reproducible report path. The final integrated dataset for the selected release is `reports/release/final_integrated_dataset.jsonl`.

8 Reproducibility and Traceability

Submission-grade path:

```
uv sync --dev --python 3.12
uv run mosaic reproduce --fixture
uv run mosaic experiment release --live
uv run mosaic experiment deterministic-scale
uv run mosaic report build
```

Full regeneration and checks:

```
uv sync --dev --python 3.12
uv run ruff check .
uv run mypy
uv run pytest
uv run mosaic reproduce --fixture
uv run mosaic report build --fixture
uv run mosaic experiment release --live
uv run mosaic experiment deterministic-scale
uv run mosaic report build
```

Requirement	Artifact	Evidence
Baseline and assisted runs	reports/release/m4_release_manifest.json	Subset A0, C-LLM, B-All, ablations, budgets
Deterministic scale runs	reports/release/tables/deterministic_scale.csv	Full Camera, Monitor, and Notebook A0 runs
Component metrics	reports/release/tables/metrics_summary.csv	Schema, blocking, linkage, clustering, fusion
Operational metrics	reports/release/tables/operational_metrics.csv	Calls, tokens, cost, fallbacks, invalid outputs
Concrete error cases	reports/appendix/m4_error_cases.json	Source records, system outputs, expected values
Final dataset	reports/release/final_integrated_dataset.jsonl	Integrated entity JSONL export

Table 12: Traceability from assignment requirements to release artifacts.

Fixture mode proves that report mechanics can be regenerated without spending money or calling external services. Subset-live mode proves that reported LLM-assisted results came from the selected dataset, committed prompts, committed model settings, and logged responses. The report tables are regenerated from manifests each time, so stale hand-entered results cannot silently survive a pipeline change.